

Homework (Jalapeno)

- Q1.** In ancient Egypt, the ratio of a pyramid's base to its height was 3 : 5. If the height of the pyramid is 150 meters, what is the length of its base?
(A) 50 meters
(B) 90 meters
(C) 100 meters
(D) 120 meters
(E) 150 meters
- Q2.** A recipe for ancient Greek ambrosia requires a ratio of 2 : 3 of honey to olive oil. If 240 units of olive oil are used, how many units of honey are needed?
(A) 120 units
(B) 160 units
(C) 180 units
(D) 200 units
(E) 240 units
- Q3.** The ancient Babylonians used a sexagesimal number system with a ratio of 6 : 10 for the number of degrees in a circle to the number of fingers on two hands. If a circle has 360 degrees, what is the ratio of degrees to fingers?
(A) 6 : 5
(B) 6 : 10
(C) 12 : 10
(D) 18 : 10
(E) 36 : 10
- Q4.** In the construction of the ancient Roman Colosseum, the ratio of the length to the width of the arena was 3 : 2. If the length of the arena was 180 meters, what was the width?
(A) 100 meters
(B) 120 meters
(C) 140 meters
(D) 160 meters
(E) 180 meters
- Q5.** A museum has a collection of ancient artifacts with a ratio of 5 : 8 of pottery to sculptures. If there are 80 sculptures, how many pieces of pottery are there?
(A) 30 pieces
(B) 40 pieces
(C) 50 pieces
(D) 60 pieces
(E) 70 pieces
- Q6.** The ancient Chinese used a ratio of 4 : 5 for the number of strings on a guqin to the number of elements in the universe. If a guqin has 25 strings, what is the number of elements?
(A) 20 elements
(B) 25 elements
(C) 30 elements
(D) 31.25 elements
(E) 40 elements
- Q7.** In ancient Mesopotamia, the ratio of the area of a sheep's pasture to the area of a goat's pasture was 3 : 4. If the area of the sheep's pasture is 240 square kilometers, what is the area of the goat's pasture?
(A) 240 square kilometers
(B) 320 square kilometers
(C) 360 square kilometers
(D) 400 square kilometers
(E) 480 square kilometers
- Q8.** The ancient Egyptians used a ratio of 1 : 1.618 for the construction of the golden rectangle. If the length of the rectangle is 100 meters, what is the width?
(A) 50 meters
(B) 61.8 meters
(C) 80 meters
(D) 100 meters
(E) 120 meters

Open Ended Questions (FRQ)

- Q9.** The ancient Greeks used a ratio of 1 : 1 : $\sqrt{2}$ for the construction of the icosahedron. If the length of one side of the icosahedron is 10 meters, find the length of the other side and show your work.
- Q10.** A historian is studying the ratio of the number of soldiers to the number of civilians in ancient armies. If the ratio of soldiers to civilians in the ancient Roman army was 3 : 7 and there were 21000 soldiers, find the number of civilians and show your work.

Chef's Tips

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- Tip 1: Make sure students understand the concept of equivalent ratios
- Tip 2: Emphasize the importance of using cross-multiplication to solve proportions
- Tip 3: Encourage students to use real-world examples to model proportions